# 03 21826 Autumn 2025
# LH The Life and Death of Stars

## Lecture 21

(JCD 16)

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# Previous lecture

- Massive star evolution creates onion-like structure in star
- Iron core will collapse in free fall after exceeding the Chandrasekhar mass
- Neutrons form in the core and rising density will stop the collapse
- A shock wave travels through the star and makes it explode, leaving a neutron star
- Supermassive stars will end with a black hole

# Slido

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# Outline

End states of stars

Observations

Summary

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# Four end states of stars

- White Dwarf: end state for low-mass and moderate-mass stars (lower than 8-10 Solar masses)
- Neutron Star: end state after a core-collapse supernova (type Ib/c or II), for massive stars
- Black Hole: end state of
  - supermassive stars (higher than 20 Solar masses)
  - neutron stars who accreted too much material
- "Nothing": end state of white dwarf that accreted too much material after which it explodes through a Supernova type Ia

# Degenerate matter

- Both White Dwarfs and Neutron Stars consist of degenerate matter
  - White Dwarfs: degenerate electrons
  - Neutron Stars: degenerate neutrons
- Need to distinguish between non-relativistic and relativistic degeneracy pressure
  - Non-relativistic:
$$P_{\rm deg} = \frac{2}{3} u_{\rm deg} \propto \rho^{5/3}$$
  - Relativistic:
$$P_{\rm deg} = \frac{1}{3} u_{\rm deg} \propto \rho^{4/3}$$
- This means pressures are in polytropic form [see lecture 6]

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# Mass-radius relation

- From the polytropic equations, we can now show that for non-relativistic white dwarfs and neutron stars

$$M \propto R^{-3}$$

- For each, the proportionality constant will be different
- Note that you can show this relation also by balancing the pressures from hydrostatic equilibrium and from the degenerate matter (verify this!)
- The mass-radius relation shows that more massive white dwarfs or neutron stars are actually more compact and would thus be denser

# Chandrasekhar limit

- From the polytropic equations, we can also show that for relativistic white dwarfs and neutron stars mass is independent of radius and we derive a fixed mass.
- This mass is called the Chandrasekhar limit. It is an upper limit.
  - White dwarfs ($A$ and $Z$ depend on the nuclei):

$$M_{\mathrm{Ch}} = 1.457 \left(\frac{2Z}{A}\right)^2 \mathrm{M}_\odot$$

  - Neutron stars:

$$M_{\mathrm{Ch}} = 5.73 \mathrm{M}_\odot$$

- Note that, again, you can show that there must be a mass limit from balancing the pressures from hydrostatic equilibrium and from the degenerate matter (verify this!)
- Both this limiting mass discovery (Chandrasekhar 1931) and the discovery of the neutron and then neutron stars (Chadwick 1932) were separately awarded with a Nobel Prize in Physics

# The physics of black holes

- Requires general relativity and quantum-mechanical effects of gravity to fully describe
- In short: the gravitational field in the collapse is so strong no other forces can resist it.
- Matter is "crushed out of existence"
- A singularity is formed [within the framework of classical physics]

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# Observations of white dwarfs and neutron stars

- After these stars are formed, they are still quite hot so while they are not generating nuclear energy, they do have a reservoir of thermal energy for a long time
- We can observe this thermal radiation [luminosity]
  - Lots of observations for white dwarfs by now - confirming the mass-radius relation
  - Neutron stars much harder to observe thermally but getting there
- Neutron stars can be observed through other parts of electromagnetic radiation or indirectly:
  - Radio pulses - pulsars are highly magnetised neutron stars with extremely precise pulses
  - X-rays
  - Gravitational Waves

# White dwarfs through Gaia

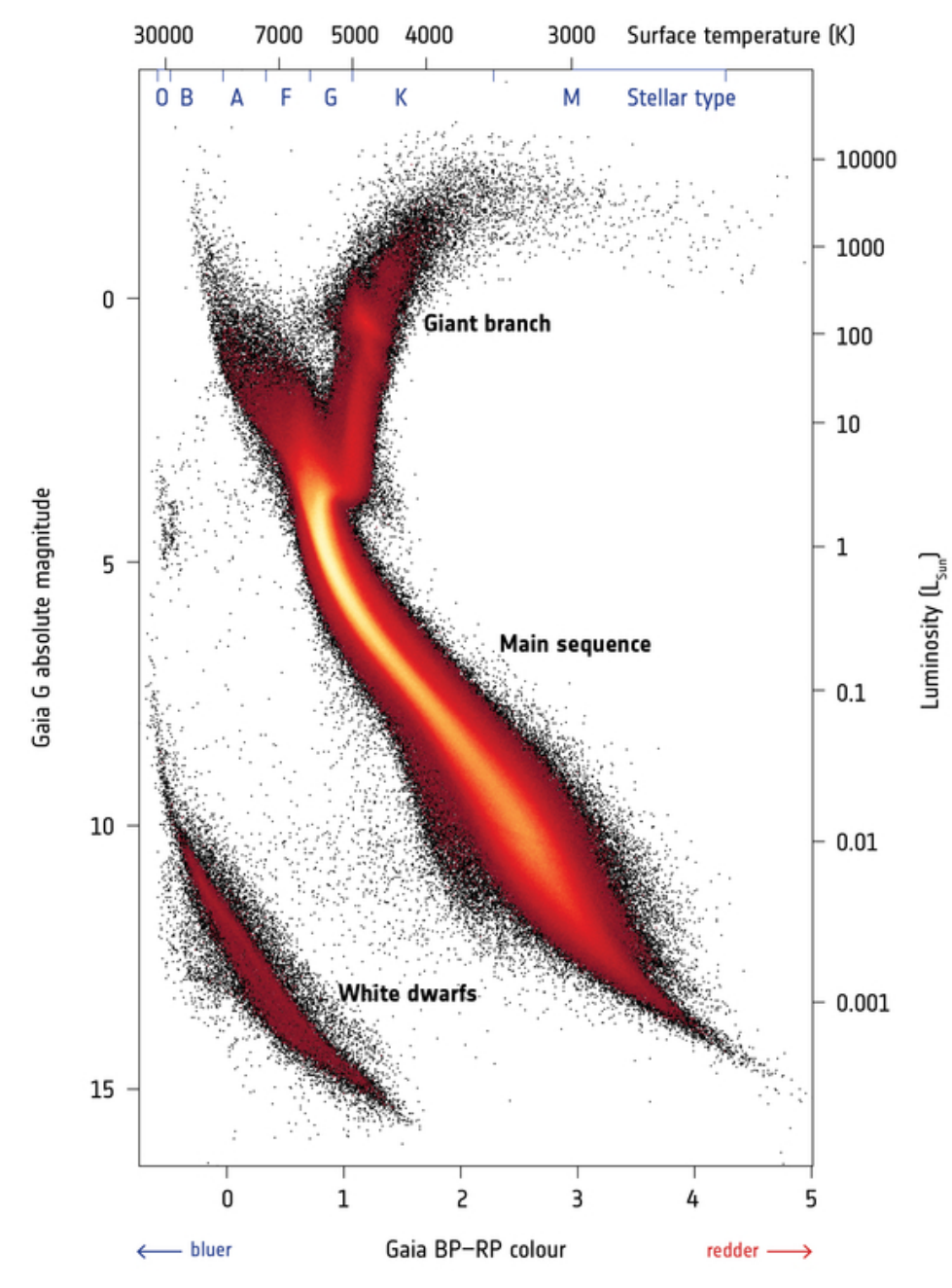


Image credit: ESA

# Neutron star imaging and spectroscopy

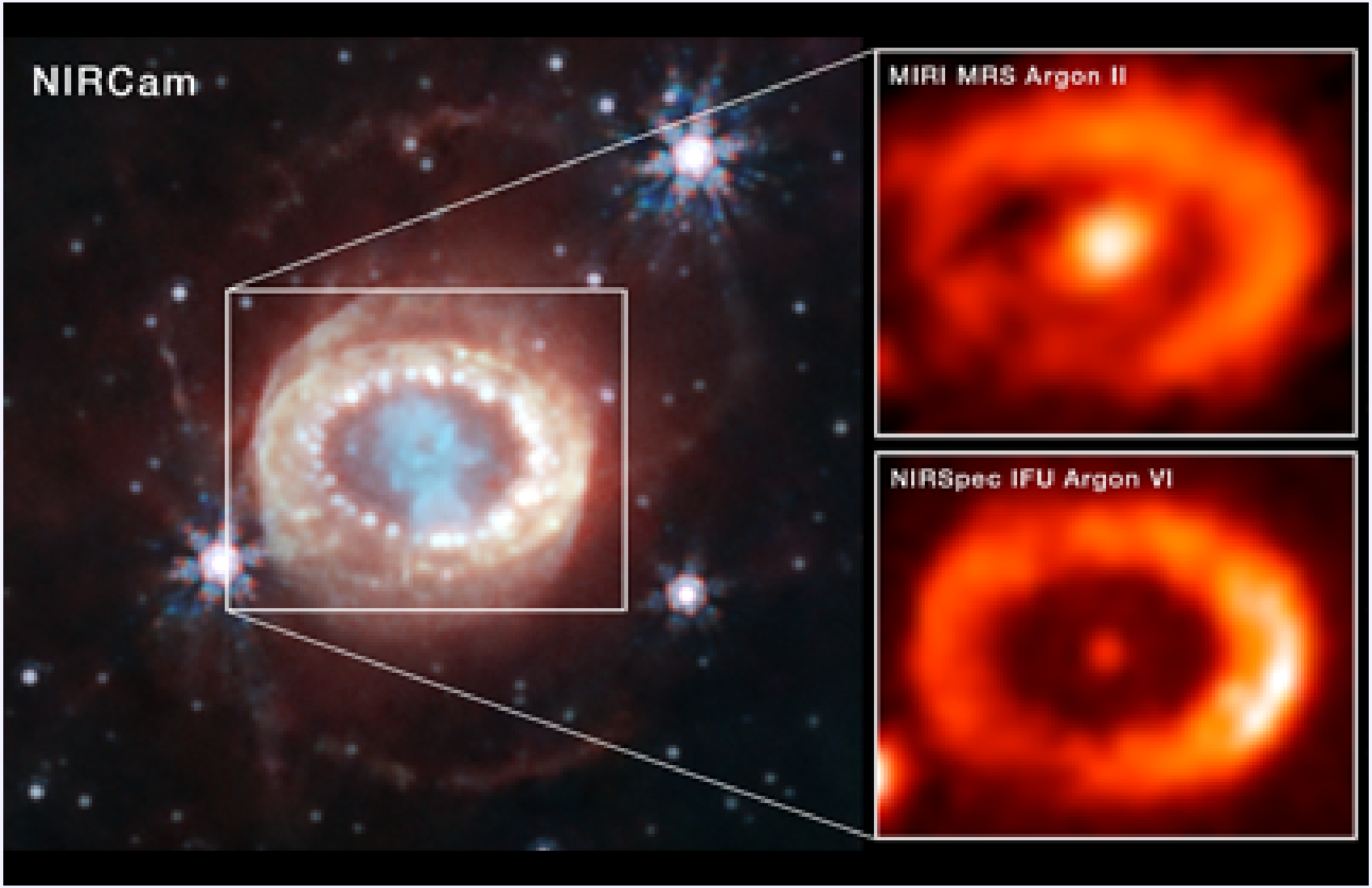


Image credit: Fransson et al. 2024

# Jocelyn Bell Burnell discovered pulsars as a PhD student! but her supervisor got the Nobel Prize alone

Image credit: University of Oxford

The Bell Burnell foundation now funds students from underrepresented groups to pursue a PhD in (astro)physics

# Observations of black holes

- Black Holes do not emit light themselves - hence the name
- Black Holes have been indirectly observed:
  - Through observations of stellar motion near Black Holes, typically in galaxy centres [including the Milky Way]
  - Through the accretion disc formed around the Black Hole, created from the infalling matter. The disc can emit X-rays or radio waves
  - Through gravitational waves

# Black Hole Observation

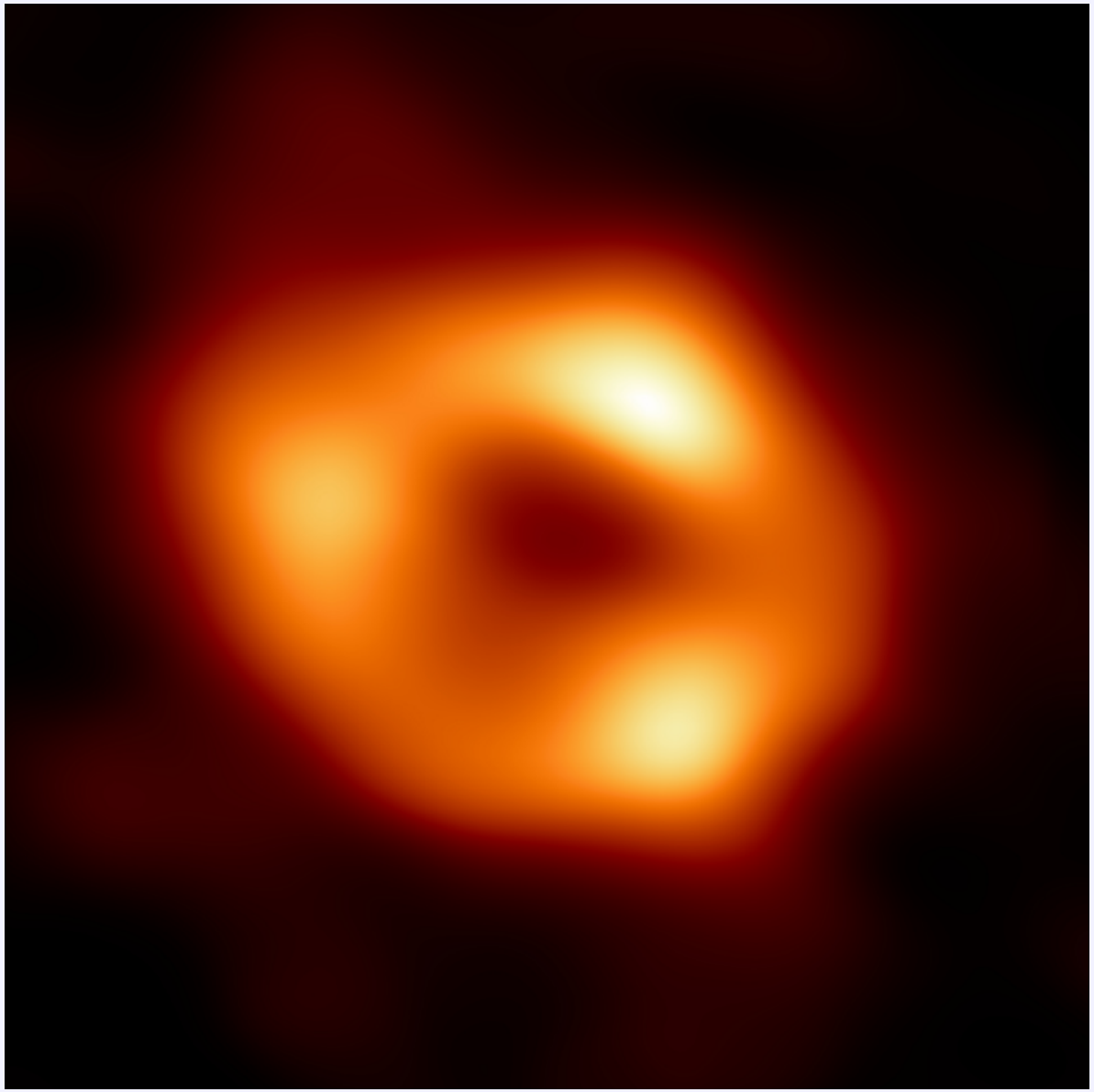

Image credit: Event Horizon Telescope Collaboration 2022

# Summary

- Stars can end up as white dwarfs, neutron stars, black holes, or nothing (after a supernova type Ia)
- White dwarfs and neutron stars can be described as polytropes with degeneracy pressures
- The Chandrasekhar limit is an upper limit for both
- The more massive they are, the more compact they are

NEXT LECTURE: Summary / Revision lecture